

PORT MEADVILLE, PA, USA

WMO: 725104

Lat: 41.626N

Lon: 80.215W

Elev: 427

StdP: 96.30

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-14.0	-21.0	0.6	-14.8	-18.4	0.8	-12.9	10.1	-0.4	9.0	-2.6	3.3	210	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.8	22.0	28.5	21.4	27.4	20.8	23.7	27.7	22.7	26.5	21.9	25.5	3.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.0	26.1	21.5	17.1	24.9	20.9	16.4	24.2	72.9	28.0	69.1	26.4	65.9	25.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	-19.1	31.8	3.5	1.5	-21.6	32.9	-23.6	33.8	-25.6	34.6	-28.1	35.8		
(5)			-19.4	25.5	3.4	1.6	-21.8	26.6	-23.8	27.6	-25.7	28.5	-28.1	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.8	-2.7	1.9	9.0	14.9	19.1	21.5	20.6	17.3	10.9	5.1	-0.7	(6)
(7)		DBStd	10.23	6.51	6.22	6.41	5.53	4.82	3.65	2.90	2.75	3.90	4.88	5.25	5.43	(7)
(8)		HDD10.0	1689	428	357	264	84	12	0	0	0	1	48	163	332	(8)
(9)		HDD18.3	3532	684	589	510	284	128	34	6	11	64	234	398	589	(9)
(10)		CDD10.0	1508	2	2	13	53	164	273	357	328	222	77	16	2	(10)
(11)		CDD18.3	306	0	0	0	3	22	58	104	80	34	5	0	0	(11)
(12)		CDH23.3	1922	0	0	2	26	157	376	685	448	208	20	0	0	(12)
(13)		CDH26.7	399	0	0	0	2	23	77	167	85	43	2	0	0	(13)
(14)	Wind	WSAvg	3.0	3.7	3.5	3.4	3.4	2.8	2.5	2.3	2.2	2.3	2.9	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	1122	77	64	81	93	96	113	112	99	111	99	88	90	(15)
(16)		PrecMax	1511	147	144	157	223	185	217	246	186	207	177	128	181	(16)
(17)		PrecMin	724	36	26	33	46	33	22	23	49	35	50	24	35	(17)
(18)		PrecStd	175	28	28	31	36	41	45	58	32	46	30	27	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	16.9	22.0	26.3	29.2	30.2	32.0	31.1	30.2	25.9	20.4	15.8	(19)
(20)			MCWB	11.8	12.0	14.4	16.9	20.3	21.7	23.0	23.5	21.7	19.5	14.0	12.7	(20)
(21)		2%	DB	12.2	12.4	17.7	23.3	27.0	28.8	30.0	28.8	27.8	22.9	17.4	12.3	(21)
(22)			MCWB	10.7	9.8	12.1	15.1	19.2	21.3	22.4	21.8	20.8	17.2	13.0	10.1	(22)
(23)		5%	DB	8.6	9.2	14.8	21.0	25.1	27.4	28.5	27.4	25.9	20.9	15.3	9.9	(23)
(24)			MCWB	6.6	6.6	10.3	13.7	18.1	21.1	21.6	20.9	19.8	16.5	11.2	7.9	(24)
(25)		10%	DB	5.3	6.2	12.1	17.9	22.9	25.7	27.2	26.0	23.7	18.5	13.0	7.2	(25)
(26)			MCWB	3.3	3.3	8.3	12.0	16.7	19.5	21.0	20.0	18.7	15.0	9.6	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.3	15.3	18.9	22.1	24.2	25.3	24.3	23.9	21.1	16.0	13.1	(27)
(28)			MCD B	14.1	15.9	20.6	23.7	26.7	27.2	30.3	28.9	27.9	24.5	18.8	14.7	(28)
(29)		2%	WB	10.6	10.0	13.3	16.4	20.6	22.7	23.9	23.1	22.0	19.2	13.9	10.6	(29)
(30)			MCD B	12.3	12.2	16.5	21.3	25.3	27.0	28.1	26.7	25.6	21.4	16.2	12.0	(30)
(31)		5%	WB	7.0	7.0	11.3	14.8	19.3	21.7	22.9	22.1	20.8	17.2	12.2	7.8	(31)
(32)			MCD B	8.5	9.3	14.2	19.4	23.6	26.1	26.6	25.5	24.0	19.7	14.6	9.1	(32)
(33)		10%	WB	3.4	3.5	8.3	13.0	18.0	20.7	21.9	21.3	19.7	15.3	10.2	5.5	(33)
(34)			MCD B	5.1	5.8	11.4	17.2	21.7	24.5	25.2	24.4	22.8	18.1	13.0	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	7.7	8.6	10.3	10.3	9.5	9.4	9.2	9.3	8.4	7.3	5.9	(35)
(36)			MCD BR	10.0	11.0	12.1	13.4	11.8	10.7	10.5	10.3	11.0	10.9	10.3	9.1	(36)
(37)			MCWBR	8.8	9.1	8.1	8.2	6.2	5.8	5.0	5.1	5.4	6.3	7.4	8.2	(37)
(38)		5% WB	MCD BR	9.6	10.8	11.1	11.7	10.4	9.6	9.0	8.7	9.2	9.4	9.4	9.1	(38)
(39)			MCWBR	9.2	9.6	8.2	8.2	6.4	5.8	5.0	5.0	5.2	6.6	7.8	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.327	0.342	0.371	0.408	0.433	0.432	0.420	0.375	0.352	0.310	0.293	(40)	
(41)		taud	2.422	2.312	2.359	2.336	2.250	2.214	2.235	2.261	2.362	2.430	2.525	2.494	(41)	
(42)		Ebn at Noon	855	874	905	898	869	845	841	841	858	838	838	834	(42)	
(43)		Edh at Noon	78	102	110	121	135	141	136	129	109	90	71	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.39	2.22	3.22	4.41	5.25	5.75	5.81	5.17	4.14	2.63	1.70	1.06	(44)	
(45)		RadStd	0.12	0.26	0.36	0.47	0.49	0.48	0.45	0.28	0.41	0.26	0.24	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98	N/A

Nomenclature: See separate page